

PROJECT PROPOSAL

Course: Digital Engineering

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Group: Group U

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PATIENT SCHEDULER SYSTEM

1. PROBLEM STATEMENT

Manual hospital appointment systems lead to scheduling conflicts, long waiting times, and inefficient resource utilization.

2. INTRODUCTION

To address the challenges faced in manual hospital appointment systems, Patient Scheduler provides a web-based customer portal. This system offers the facilities for patients to book their appointment with doctors online, view and manage appointments.

3. OBJECTIVES AND SCOPE

Objectives:

- Eliminate double-bookings and manual errors with automated bookings.
- Set up web enabled patient self-service.
- Enable doctors and administrators to manage schedules and view analytics.

Scope:

- Patient registration and login.
- Doctor search by specialty.
- Appointment booking, cancellation, and history tracking.

Out of Scope:

- Billing and payment integration.
- Video consultation features.

Assumptions:

- Users have regular internet access.
- Hospital staff are trained to use the admin panel.

4. SYSTEM REQUIREMENTS

Functional Requirements:

Patient:

- Register and login
- Search doctors
- Book/cancel appointments
- View history

Doctor:

- View schedule
- Access patient details

Admin:

- Manage users
- Create reports

Non-Functional Requirements:

Responsive UI.

System response time should be less than 2 seconds for page loading.

System should ensure high availability with at least 99% uptime.

Secure authentication must be implemented to protect the data.

Technical Requirements:

Web/App Server (Java 17+).

Browsers (Chrome, Firefox).

5. SYSTEM DESIGN

Architecture:

Three-Tier MVC (View, Controller, Model)

View (Frontend):

Layouts with doctors and admins for schedules and analytics

Controller:

Responsible for receiving requests

Manage the session

Verifying inputs

Database Entities:

Patient (ID, Name, Email, Phone)

Doctor (ID, Name, Specialty)

Appointment (ID, Patient ID, Doctor ID, Date, Time, Status)

Booking Flow:

Login → Search Doctor → Check Slots → Book → Save → Notify → Update UI

6. RESULTS

Booking efficiency with an average time of one minute per appointment.

Error reduction.

Project scalability for over 1000+ users per day.

Challenges:

Addressing session timeout.

Real-time updates via polling.

7. FUTURE ENHANCEMENTS

- Migration to spring boot for REST API creation.
- Use of a mobile app created in React Native.
- Use of Artificial Intelligence to provide optimized slots.
- Implementation of a payment gate.

8. CONCLUSION

The combined effort for the development of the Patient Scheduler system signifies a strategic approach of modernizing hospital management by transition from manual appointment scheduling to an automated system. This Project focused on essential functional requirements such as registration of patient securely and conflict free booking by following the three tier MVC architecture.

Aligned with the guidelines, impactful features can be implemented within the two-week time frame. These include automated conflict detection, establishment of web-enabled service portal for registrations, a centralized dashboard that allows doctors and administrators to view schedule and analytics. This project can serve as future enterprise level web solution by providing a scalable architecture.